

Sheet 1:

Task: Planning Visualisation

Author: Manveen Pham

Date: 28/09/2024

Ideas

- Global Ageing Demographic
 - Income Group
 - % of Seniors

Choropleth Map



Quantitative



% seniors



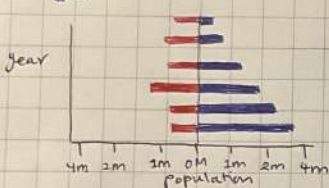
Qualitative

Income Group

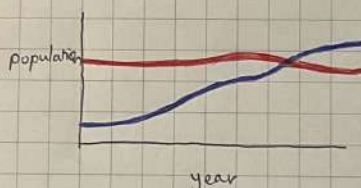
- High
- Middle
- Low

- Population Growth of older Australians

20's
65+

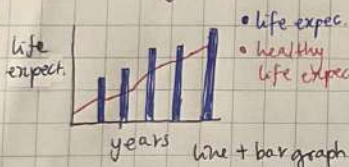


Population Pyramid

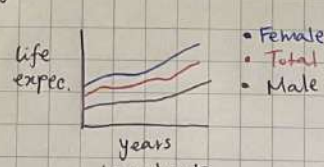


Line chart

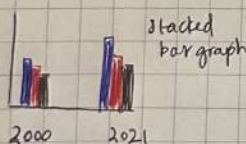
- Increasing life expectancy



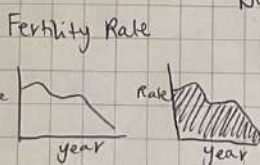
Line + bar graph



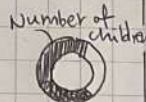
Line chart



Stacked bar graph

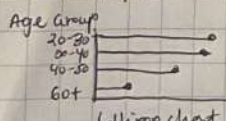


Fertility Rate

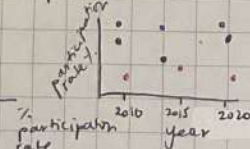


Number of children

- Workforce Participation



Lollipop chart



% participation rate

Age Group

- 20-30
- 40-50
- 65+

Filter

Population Pyramid

difficult to see a general trend over multiple years

↳ harder to annotate key events.

Fertility Rate line graph

↳ Idiom too frequently used

still want to keep trend over time.

Qualitative chart

Shifts focus away from group → Quantitative with shapes to visually indicate both % of seniors and income group?

Categorise

Global Age Demographic

- 7. seniors
- Income groups

Population Group

- 20 and under
- 65 and over

Increasing life expectancy

- life expectancy
- healthy life expect.

Workforce participation

- Age groups
- Participation Rate

Combine & Refine



+



=>



Qualitative → Symbols / shapes

Quantitative → Colour gradient

Question

1. Is this too visually cluttered?
2. What is a more effective way of enabling comparison?

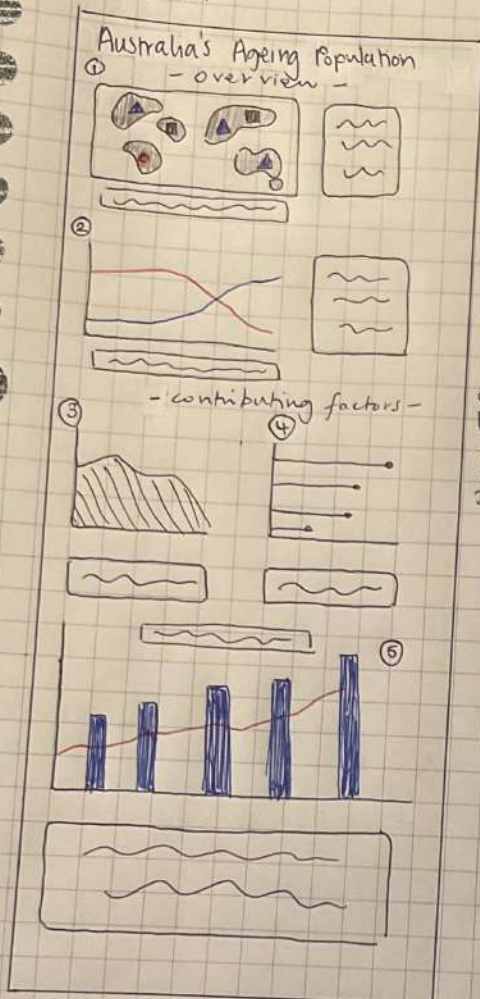
Sheet 2

Task: Designing a Dashboard

Author: Manveen Phain

Date: 28/09/2024

LAYOUT



OPERATION

Discussion

- Is ③ too simple? Maybe further interaction can be added to allow readers analyse distance b/w the 2 age groups

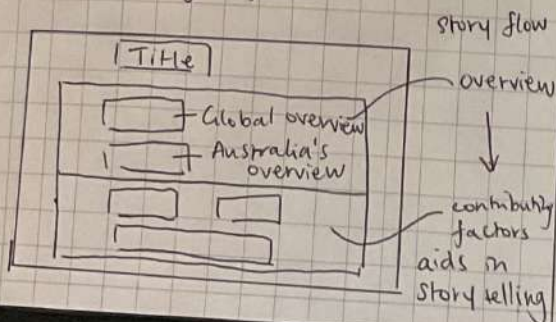
- brushing?
- tooltip?
- filtering?

↳ might not be necessary with only 2 groups

- Currently ⑤ doesn't allow for intuitive comparison b/w the two sexes, even w/ dropdown.
 - ↳ Use line chart?
 - ↳ Stacked bar chart
 - ↳ is a bit too visually dense across time periods.

Focus

- there is no single point of focus, everything is focused on.



Sheet 3

Task: Narrative Story Board

Author: Maureen Pham
Date: 28/07/2024

LAYOUT

1 Australia's Ageing Population	Australia, like many developed nations, has seen a shift towards an older population	3 Tip: Hover to view % seniors and income level
4 Australia's oldest population is rapidly increasing	5 	6 What do you think is contributing to this? ☐ Increasing life expectancy ☐ Lower fertility rates
7 One of the major contributing factors is increased life expectancy	8 Life  Health 	9 Another major contributing factor is lowered fertility rates
10 	11 A side effect of this is a reduced workforce 	

Scatterplot looks much nicer than lollipop chart

6 After submitting a mark will show up next to the text

- ☒ Increasing life expectancy ✓
- ☐ Lower fertility rates X

The correct answer is

operations

▶ Next slide button (present on all slides) which allows users to click on to next slide

Year slider that animates the movement of the line chart

Multiple choice options to introduce upcoming contributing factors

- ☐ } multi-select boxes
- ☐ } for users to check

submit button to validate chosen options

Focus: Even focus on both text and charts → the text leads the story telling and the charts keep the users engaged

In chart sections, there will be a small text box providing hints on how to navigate the charts.

Discussion

- Slides with just text might be too wordy.
↳ Use images / chartjunk to keep readers engaged?
- Might need to keep text brief - however can result in less content if so
- Can this be implemented with Javascript?

Sheet 4

Task: Scrollable Poster with Narration

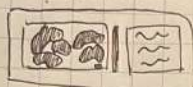
Author:

Maureen Pham

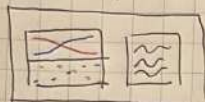
Date: 28/09/2024

LAYOUT

Australia's Ageing Population
Global overview



Australia's Overview



Contributing factors

Increasing life
Expectancy



Lowering
Fertility Rates



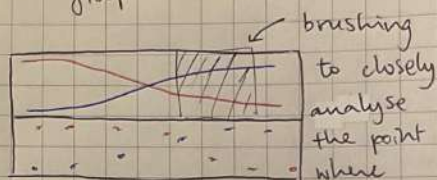
Focus

focus is on everything, however there is more emphasis on the charts and the text is more supplementary to aid story telling and to provide context.

OPERATIONS

year slider to animate movement of line chart

↳ add another button that animates both graphs simultaneously?



↑ sections overtake those aged 20 and under
can add a Rule to clearly annotate the point.

Discussion

- How can I make this more visually appealing?

↳ stack all elements vertically? Consistent line of sight / don't have to worry about adjusting for different chart + text sizes.

↳ Add more hierarchical elements

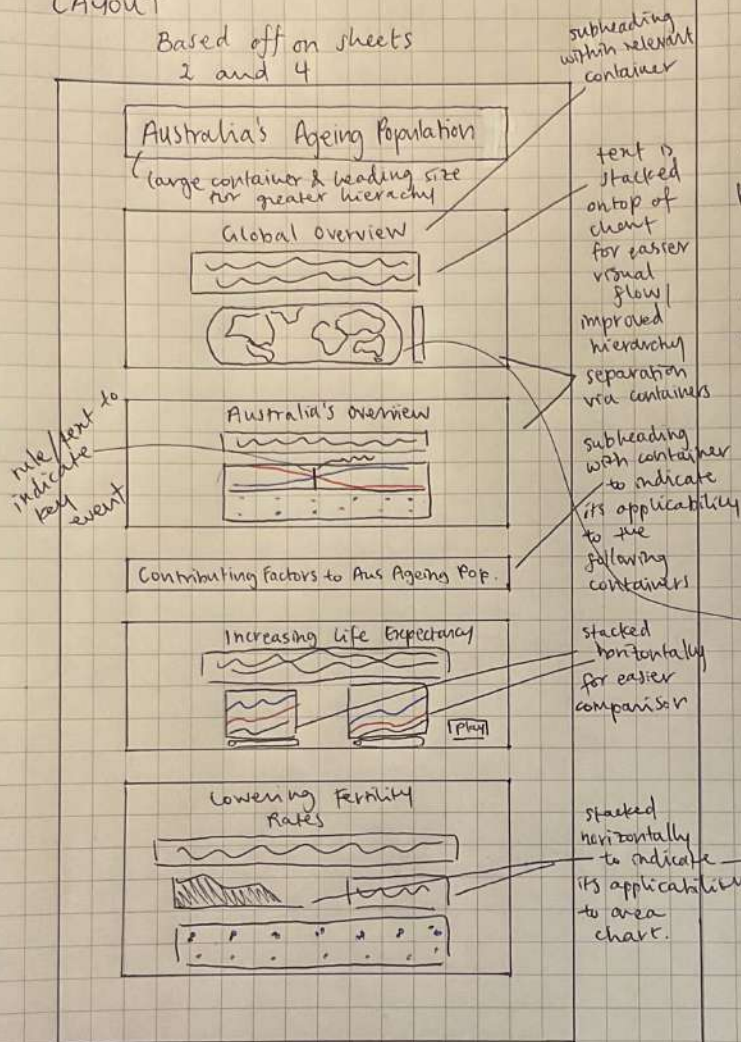
Sheet 5

Author: Manveen Pham

Task: Designing the Final Layout Date: 28/09/2024

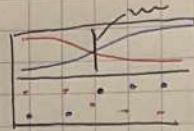
LAYOUT

Based off on sheets
2 and 4



OPERATIONS

Play button to simultaneously animate both line charts, along with year slider to give users the choice of seeing change from both charts or 1.



brushing to allow users to more closely see points / distance

- tooltip -

Country:	—
Income:	—
% seniors:	—

Details

Updated values / animation implemented using JavaScript

- JS, VegaLite

Need topojson files for map (ocean, graticule etc.)
↳ EqualEarth

Specific requirement:
VegaLite, map, laptop

time to build: 1 week 1/2.

Focus is on everything - this is achieved by separating subtopics in its own containers, and emphasising of hierarchy of elements for intuitive flow and greater storytelling